School of Computer Science and Engineering

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Meridian Data
An AI-Powered NL-to-SQL Database Explorer

Internal

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1. Title and Problem definition

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- **Problem —** querying databases requires SQL fluency, a barrier for students, analysts and domain experts.
- **Existing systems —** single proprietary engine, closed-source, and send data to the cloud unconditionally.
- They hallucinate SQL and lack safety, authorization and human oversight.
- **Novelty —** hybrid deterministic + LLM querying over eight engines, with layered safety, human review and cloud-to-local privacy.

Introduction

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- Converts plain-English requests into validated, dialect-specific SQL using LLMs given the live schema.
- **Scope —** NL querying, deterministic introspection, AI analysis with auto-charts, dashboards, and CSV / PowerPoint export.
- Connects to SQLite, PostgreSQL, MySQL, MSSQL, Oracle, MongoDB, Cassandra and Redis via one adapter layer.
- **Doubles as a productivity tool and a DBMS teaching aid —** it shows and explains the SQL it generates.

Objectives

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- Convert natural language into validated, dialect-aware SQL via a schema-aware prompt.
- Answer common introspection commands deterministically, without the LLM.
- Enforce two-layer safety with role-based access control.
- **Keep a human in the loop for writes —** dry-run, snapshot and rollback.
- Provide AI analysis, auto chart selection, dashboards and export.
- Support cloud (Groq) and local (Ollama) models with automatic failover.

Literature Survey

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Evolved from rule-based NLIDBs to neural text-to-SQL (Seq2SQL / WikiSQL, SQLNet) and the Spider benchmark.

RAT-SQL and BRIDGE solved schema linking; PICARD added constrained decoding for valid SQL.

LLM era — in-context learning, DIN-SQL decomposition, DAIL-SQL prompting; BIRD shows real databases stay hard (GPT-4 ~55%).

Safety / RBAC and human-in-the-loop research inform the design.

Meridian Data integrates these validated components rather than proposing a new parser.

Literature Survey — Summary

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Work	Technique	Key Limitation
Seq2SQL (2017)	RL over WikiSQL	Single-table queries only
SQLNet (2017)	Sketch + column attention	Tied to WikiSQL grammar
Spider (2018)	Cross-domain benchmark	Schema-only; clean DBs
PICARD (2021)	Constrained decoding	Syntactic, not semantic, validity
DIN-SQL (2023)	In-context decomposition	High token cost / latency
BIRD (2023)	Dirty-DB benchmark	GPT-4 only ~55% accuracy

Proposed Methodology

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Intent classification — introspection answered deterministically; data questions go to the LLM.
Schema-aware generation — schema (columns, keys, indexes, samples) injected into a dialect prompt; Groq / Ollama failover.
Validation & authorization — classify (READ / WRITE / SCHEMA / SYSTEM) → safety filter → role check.
Execution or review — READ runs paginated; WRITE / SCHEMA go to human review with snapshot.
Analysis & presentation — AI summary, auto-selected chart, CSV / PowerPoint export.

Proposed Methodology — Pipeline

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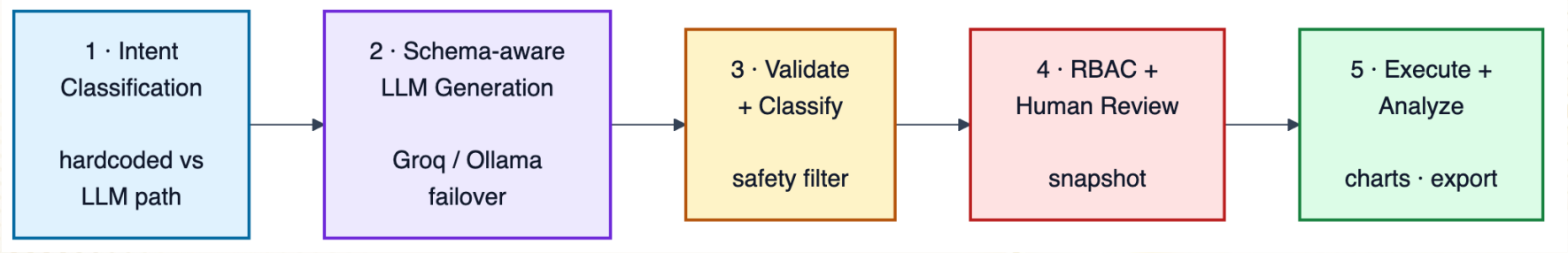

Requirement Specification

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Functional — NL→SQL generation; deterministic introspection; classification + safety; RBAC; human review with dry-run / snapshot; multi-DB connect; AI analysis, dashboards and export.

Non-functional — security: block DDL / injection, encrypt credentials (Fernet).

Reliability — introspection always works; automatic cloud↔local failover.

Performance — millisecond introspection, sub-second cloud generation.

Usability, privacy (local mode) and extensibility (new engine = new adapter).

Design approach

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Layered — presentation → application engines → LLM provider (failover) → data-access (Adapter) → persistence.

Adapter pattern — one DatabaseAdapter interface (connect, get_schema, list_tables, execute) with eight concrete adapters; app code never imports a driver.

A two-layer guardrail (classification + safety) plus an explicit human gate isolate the LLM from the database.

Conversation context (last five turns) enables follow-up and query refinement.

Architecture diagram

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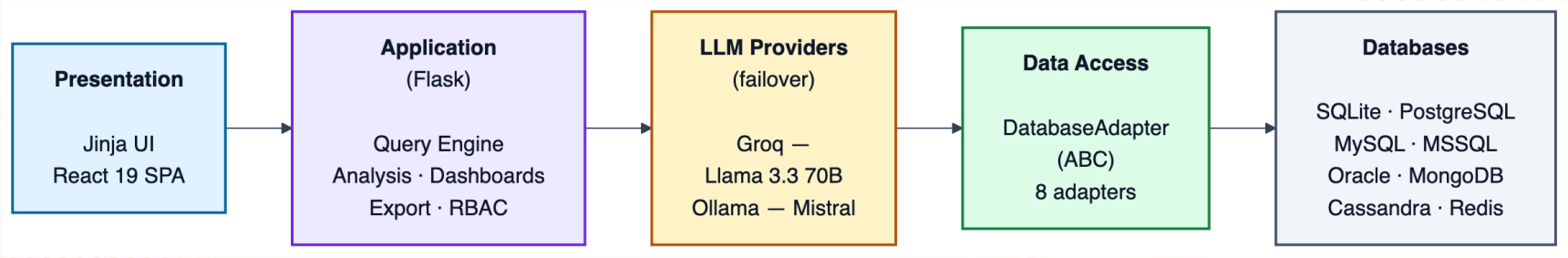

Database Design — ER Diagram (Sample Schema)

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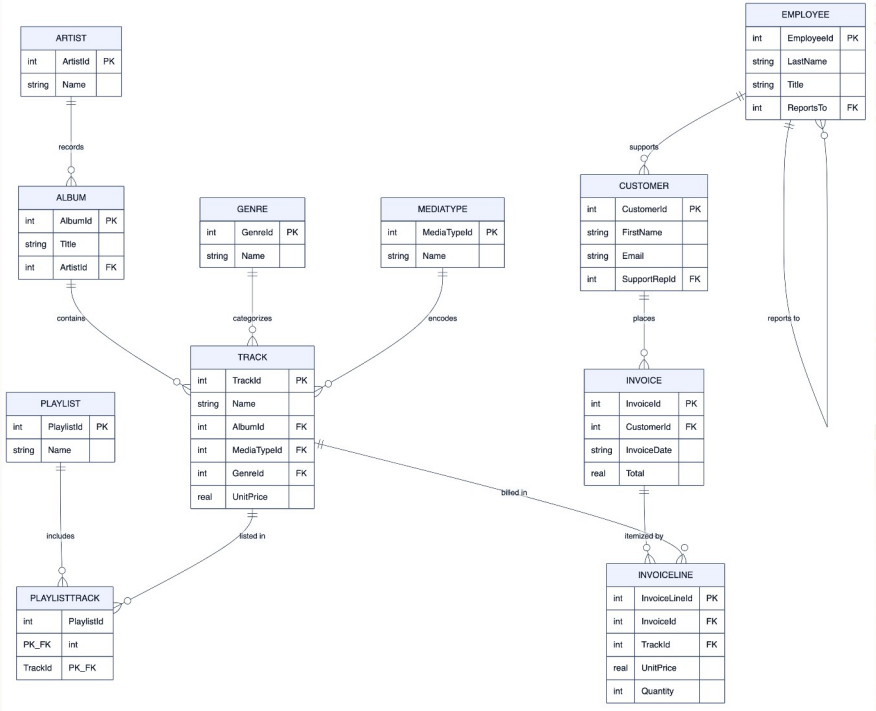

Innovations

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- **Hybrid execution —** deterministic introspection + LLM generation: correctness where cheap, intelligence where needed.
- One adapter abstraction extends NL querying across eight relational and NoSQL engines (JSON contract for Mongo / Redis).
- **Cloud-to-local failover —** resilient and privacy-flexible.
- **Defence-in-depth —** the LLM is never trusted to be safe; a separate validator and human gate guard execution.

Implementation

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- Python 3.11 + Flask backend; two front-ends (Jinja + React 19 SPA) over one JSON API.
- **LLM layer —** Groq SDK (Llama 3.3 70B) and local Ollama (Mistral) with per-dialect prompt templates.
- Validator, Fernet-encrypted connection manager, and a per-engine snapshot / rollback engine.
- AI analysis via Groq JSON mode; dashboards persisted as JSON; PowerPoint export via python-pptx.

Implementation — Working Application

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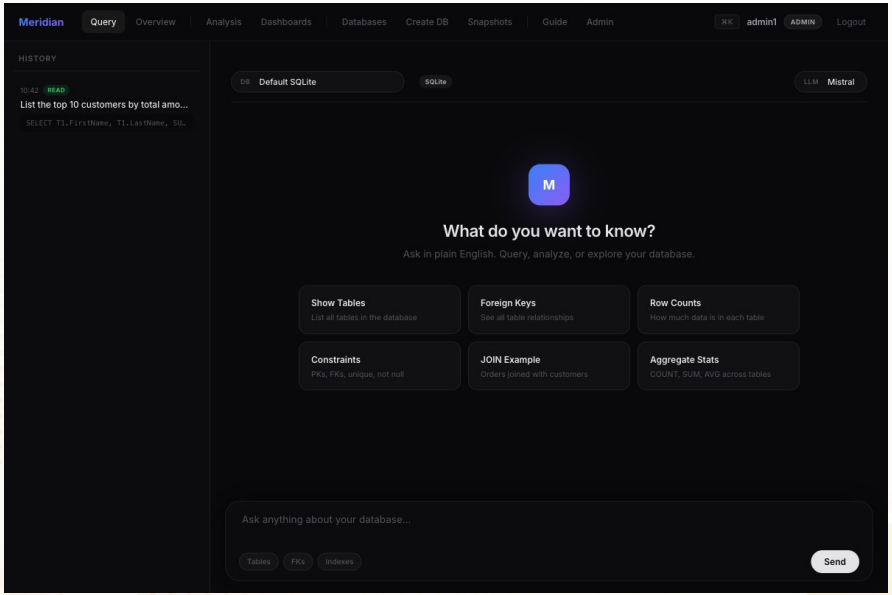

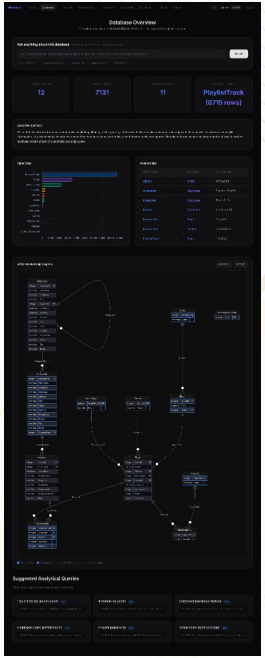

Evaluation Metrics

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Functional correctness — each feature works on the sample databases.

Generation correctness — generated SQL executes and returns the intended rows.

Safety — destructive and unauthorized statements are blocked.

Latency — time to answer, by execution path and provider.

Token efficiency — prompt and completion tokens per call.

Testing & Validation

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Unit — hardcoded commands (show tables = 12, describe, foreign keys) and the validator: all pass in < 20 ms.

Integration — generation → validation → authorization → execution; write-review → snapshot → execute; failover: all pass.

System — end-to-end NL query, DB switching, AI analysis, dashboards, destructive-command blocking: all pass on both front-ends.

Performance Analysis

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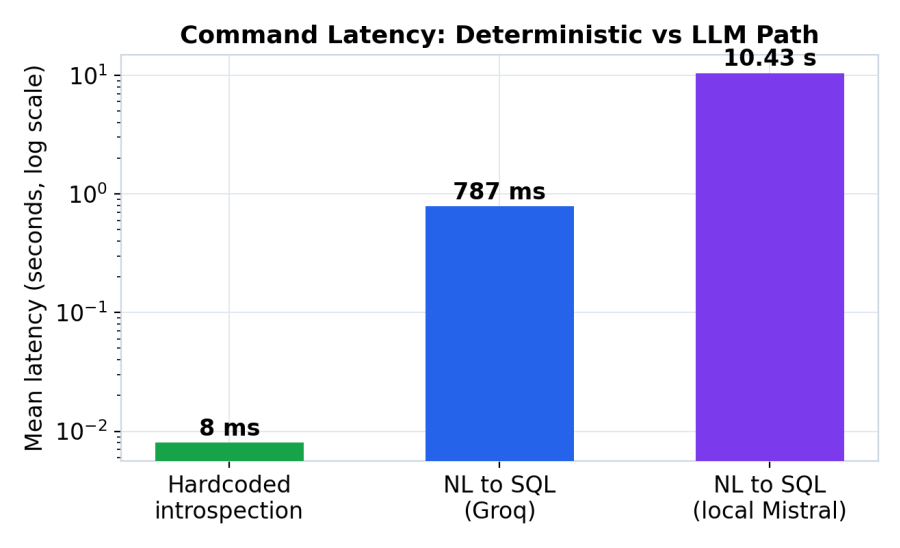
Groq cloud NL→SQL — mean 0.79 s, median 0.73 s (46 logged calls).

Local Mistral — ~10.4 s (fully private, on-device).

Hardcoded introspection — ~8 ms end-to-end (no LLM), ~100× faster.

Generated SQL validated against expected results on the Chinook DB.



Evaluation Results

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100%	0.79s	≈8 ms	3/3
Unsafe queries contained (12/12)	Mean NL→SQL latency (Groq cloud)	Hardcoded introspection (end-to-end, no LLM)	Legit reads allowed (no false blocks)

Method — labelled NL→SQL queries on the Chinook DB (generated SQL executed and compared to gold result sets), a 12-prompt destructive / injection battery, and latency over 113 logged LLM calls.

Result — correct SQL across aggregate, projection, top-N, group-by and join queries; 100% of unsafe statements contained (10 blocked, 2 routed to human review) with no false blocks on valid reads.

Hardware and/ or System Requirement

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Software — Python 3.11+ & Flask; Groq SDK / Ollama; DB drivers (psycopg2, PyMySQL, pymongo, redis, ...); cryptography; python-pptx; React / Vite / Tailwind.

Hardware — any modern browser; 4 GB RAM (cloud) or 16 GB + optional GPU (local); 2–10 GB storage.

Why — Flask is lightweight; Groq's LPU gives low latency; Ollama enables privacy; per-engine drivers give multi-database reach.

Conclusion

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Meridian Data turns LLM text-to-SQL into a dependable tool — schema-aware prompting, deterministic shortcuts, layered validation, RBAC, human review and provider failover.

Achieves sub-second cloud generation and millisecond introspection, and blocks every destructive operation.

Serves as both an analyst productivity aid and a DBMS teaching instrument.

Future Work

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Replace the keyword safety filter with a full SQL parser (e.g. sqlglot).
Unify the duplicated Jinja and JSON pipelines behind one service layer.
Add retrieval-augmented schema selection for very large schemas.
Self-correction — feed execution errors back to the model to repair queries.
Stronger credential security; a quantitative accuracy study on Spider and BIRD.

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Thank You